



Buffer Stock Saving with Heterogeneous Agents^{*}

An example of the Econ-ARK MyST template

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Abstract We solve a buffer stock saving model with the endogenous grid method and show how the Econ-ARK template typesets a working paper: title block, margin notes, equations, tables, code and author-year citations.

Keywords consumption, precautionary saving, heterogeneous agents

JEL codes C63, D14, E21



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Code

github.com/econ-ark/econ-ark-myst

Run online

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1 Introduction

Precautionary saving arises when income risk interacts with a convex marginal utility (Carroll 1997). The endogenous grid method (Carroll 2006) makes such models cheap to solve, because it replaces root finding with an inversion of the Euler equation.¹ Households hold a buffer of wealth against bad income draws. The size of that buffer depends on patience, the return on saving and the degree of risk.

2 Model

The consumer maximizes expected discounted utility subject to the budget constraint

$$v(m_t) = \max_{c_t} u(c_t) + \beta \mathbb{E}_t[v(m_{t+1})], \quad m_{t+1} = R(m_t - c_t) + y_{t+1}. \quad (1)$$

The Euler equation implied by (1) is $u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})]$, and with constant relative risk aversion $u(c) = c^{1-\rho}/(1-\rho)$.

Working paper

September 16, 2026

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^{*}Prepared for the Econ-ARK template documentation. Jane Economist: Support from the Alfred P. Sloan Foundation is gratefully acknowledged.

¹The method inverts the first-order condition on a grid of end-of-period assets rather than solving it on a grid of market resources.

2.1 Calibration

Table 1 Baseline calibration.

Parameter	Description	Value
β	Discount factor	0.96
R	Gross interest factor	1.03
ρ	Relative risk aversion	2.00

3 Results

Table 1 lists the parameters used in the solution. The consumption function is concave. Its slope falls toward the perfect foresight marginal propensity to consume as wealth grows.

```
from HARK.ConsumptionSaving.ConsIndShockModel import
IndShockConsumerType

agent = IndShockConsumerType()
agent.solve()
```

Acknowledgments

We thank the Econ-ARK community for comments.

Data availability

All code needed to reproduce the results is in the repository linked in the margin.

Declaration of competing interest

The authors report no competing interests.

Declaration of generative AI use

The authors used a large language model to draft the example text of this template and reviewed every sentence before release.

References

- Carroll, Christopher D. 1997. "Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis." *The Quarterly Journal of Economics* 112 (1): 1–56. <https://doi.org/10.1162/003355397555109>.
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